

Readying Linux

for Today's Embedded Devices

Cutting down Linux's boot time is one last hurdle that needs to be conquered. Work on that front is going on...

Linux is making rapid strides into the embedded world. There are several factors that are driving this trend. Many devices, like netbooks, set-top boxes, mobile devices, assorted gadgets, media players, etc., are Linux powered.

This article looks at why Linux is popular in the embedded world, the problem of long boot times, and the work being done to address that.

What drives embedded Linux

Till some years ago, proprietary operating systems ruled the embedded world. Linux was popular on the server side (running the websites, enterprise applications, etc). Linux's inroads into Microsoft's turf of the desktop market hadn't been too successful and hence it was written off as a niche player, mostly in the server market.

In the last three to four years, however, there has been a change in the market. Linux is getting adopted in embedded devices and the trend is accelerating. The reasons for the

shift to Linux are many:

1. Linux as an operating system is quite mature now. The 2.6 version of the kernel is feature rich and supports kernel pre-emption and threading.
2. There's rich support for the drivers of common devices and peripherals. It's quite easy to find the drivers for WLAN, Bluetooth, infrared, I2C, GPIO, etc. This reduces development time.
3. There's a good supportive ecosystem of developers, forums, mailing lists and service companies that offer porting, performance tuning and testing services.
4. Last but not the least is the fact that the OS is completely open source and is royalty free.

